
Context-Gated Knowledge-Graph Retrieval over Land–Atmosphere Literature for Land-Use Scenario Design

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Abstract

Deciding which land-use changes to test in a convection-permitting weather model is expensive: each candidate scenario costs a full numerical experiment, so the candidate set must be narrowed before any model runs. The literature is the natural filter, but the effects it reports reverse sign with season, background wind, soil moisture, patch scale, and latitude, and retrieval systems that return passages or entity communities strip those conditions away and answer confidently without them. Our position is that the study conditions under which a finding holds must be retrievable objects in their own right, not incidental text inside a passage. We build ClimateKG, which extracts each paper in stages into a graph of StudyContexts, conditioning Facets, and directed ScientificClaims, and scores environmental applicability before traversing any mechanism edge. On a shared 18-paper corpus we benchmark it against plain passage RAG and an entity-adapted Microsoft GraphRAG under a matched generation model, context window and reasoning budget, over ten questions with claim-level gold evidence, and release a hand-annotated extraction ground truth for ten papers. Staged extraction recovers 0.62 of main findings at 0.98 precision, and what it drops is overwhelmingly the conditioning structure — diurnal sign reversals, applicability thresholds, and latitude-dependent branches — which is both the argument for the approach and its clearest current limit.

1 Introduction

Land-use and land-cover (LULC) change alters local climate through surface energy and water partitioning, and the practical question for a planning agency is not whether an effect exists but which candidate change to adopt [Pielke Sr. et al., 2007, Cao et al., 2020, Alkama and Cescatti, 2016]. Answering that credibly requires a convection-permitting model run driven by a perturbed land-surface state, at a resolution that resolves the mesoscale circulations through which land heterogeneity acts [Prein et al., 2015, Ban et al., 2021, Skamarock et al., 2019]. Such runs are expensive enough that only a handful of scenarios can be tested, so the binding constraint is scenario selection, not simulation accuracy. This paper addresses the upstream problem: using the literature to assemble a small set of physically plausible, environment-matched candidate LULC scenarios for a watershed, which a modelling group can then afford to test.

The obstacle is that the literature does not report context-free effects. Afternoon rain falls preferentially over drier soils between 12:00 and 15:00 and over wetter soils between 21:00 and 03:00 [Taylor

et al., 2012]; deforestation warms the surface in the tropics and cools it in the boreal zone [Luo et al., 2024]; a wind farm warms near-surface air under a positive lapse rate and cools it under a negative one [Baidya Roy and Traiteur, 2010]; Saharan solar farms increase rainfall at 15% panel efficiency and suppress it at 45% [Li et al., 2018]; observed deforestation in Rondônia changes rainfall in the dry season under weak synoptic wind and not at all in the rainy season [Wang et al., 2000]. Eight such reversals appear in the 18-paper corpus used here (Appendix A). An answer that omits the condition is not incomplete — it is wrong for one of the two branches.

Existing retrieval discards exactly that structure. Passage-level RAG [Lewis et al., 2020] returns chunks nearest a query embedding, so conditions survive only when they happen to sit in the retrieved chunk, and the generator has no representation separating a study’s setting from its result. Graph retrieval [Edge et al., 2024] improves multi-hop synthesis by clustering entities with hierarchical Leiden [Traag et al., 2019], but a community is a thematic neighbourhood, not a statement about where a finding applies: claims from incompatible study environments cluster whenever their vocabulary agrees. In both designs applicability is a byproduct of lexical similarity, so neither can decline to transfer a mechanism to a setting no source paper studied.

We therefore make the study environment a first-class retrievable object. ClimateKG indexes each paper in stages — a map of the settings the paper actually analyses, then conditioning Facets per setting, then directed ScientificClaims scoped to a setting or to a comparison — so every claim carries the conditions under which its authors reported it. At query time the user’s area is described in the same Facet language, enriched where possible from reanalysis and climate classification [Hersbach et al., 2020, Beck et al., 2018], and applicability is scored *before* any mechanism edge is traversed. Answers ground to claim identifiers and page-level source blocks, and the pipeline reports when spatial transfer is blocked instead of producing a placement recommendation. We contribute (i) this context-gated design, in which conditions gate traversal rather than decorate it; (ii) a controlled three-system benchmark over one corpus with matched generation settings, on ten questions with claim-level gold evidence; and (iii) a hand-annotated extraction ground truth over ten papers separating main from supporting findings, which quantifies what staged extraction recovers and what it systematically drops.

2 Method

Literature corpus. The working library holds 138 papers on LULC–climate interaction, organised by intervention class — deforestation (46), generic LULC change (28), afforestation (7), irrigation (5), wind and solar farms (5), and single or paired papers on rice paddy, reservoirs, cropping practice, plantation, greening, degradation, cropland expansion and wetland restoration — following the interventions actionable at watershed scale in the target region. All experiments use a frozen 18-paper subset spanning the mechanisms above, so every system indexes identical text; it yields 2,746 SourceBlocks.

Staged indexing. Extraction proceeds in three stages rather than one, because the object a claim belongs to must exist before the claim can attach to it (Figure 1). Documents are parsed into stable SourceBlocks carrying page and section provenance. The *paper map* stage builds a tree of the study settings the paper actually analyses — each simulation branch, season, forcing level, site, and control–treatment pair — with the comparisons between them. The *facet* stage attaches conditioning records per setting across spatial-configuration, land-surface, hydrology, atmosphere, climate, and terrain domains. Only then does the *claim* stage run, once per setting or comparison, extracting directed relationships between two endpoints with a relation type, evidence role, conditioning facets, and exact supporting SourceBlocks. Per-scope extraction is what makes conditional findings expressible: a seasonal null result and a dry-season rainfall response become separate records because they belong to separate scopes, whereas a single whole-paper pass has nowhere to put the distinction. Downstream stages are refinements — conservative state canonicalisation through an explicit alias registry, and optional spatial enrichment. The 18-paper graph holds 181 StudyContexts, 556 Facets, 58 Transitions, 279 ScientificClaims, 411 States and 1,485 evidence links.

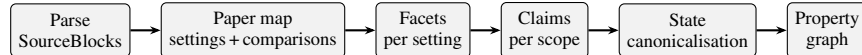


Figure 1: Staged indexing. Optional refinements — spatial enrichment feeding the facet stage, and context reconciliation feeding the claim stage — are omitted. Claims are extracted once per setting or comparison, so a finding holding only in one season or forcing regime has a scope to belong to.

Baselines. Both baselines consume the same exported SourceBlock text with references and page furniture removed. **Plain RAG** chunks it into 287 passages, embeds them, and retrieves the top eight by cosine similarity with a light metadata boost. **Entity-adapted GraphRAG** is Microsoft GraphRAG [Edge et al., 2024] with its extraction stage replaced by ours: canonical States become entities and ScientificClaims become directed relationships carrying relation type, study context, conditioning facets, paper and supporting blocks. Everything downstream — hierarchical Leiden clustering, community summarisation, community retrieval — runs unmodified, giving 399 entities, 279 relationships and 134 communities, all at level 0 because most components already fall below the maximum community size. This is deliberately not an independent extraction baseline; it isolates whether community retrieval uses the *same* extracted objects better than context-gated traversal.

Query pipelines. ClimateKG parses the question into optional source and target endpoints, an intervention description, and explicit context facets in the same schema as the index. Place names are recorded but never expanded into assumed environmental properties; those come from the user or from deterministic enrichment. Context retrieval scores each paper setting for semantic similarity and coverage, keeping missing coverage distinct from genuine mismatch. Claims are ranked over separate context, mechanism, intervention, and endpoint channels combined only at ranking time; direct evidence and graph-traversed paths form separate lanes; and the top eight paths plus contradicting alternatives are packaged with their exact SourceBlocks for synthesis. GraphRAG instead embeds the question, selects the top five communities, re-ranks their relationships, and supplies up to 18 claim records. All three systems use qwen3.6:27b for generation, qwen3-embedding:4b for embedding, a 32,768-token context, and a medium thinking budget.

3 Experiments

Ten questions are frozen in advance: eight in-corpus questions with gold ScientificClaim identifiers read off the extracted claim inventory, and two out-of-corpus transfer questions with expected source papers but no gold claims, since a correct answer there is a conditioned hypothesis rather than a retrieval hit. They probe single-paper mechanisms, thresholds, single-variable sensitivity, cross-paper agreement, mechanism chains, observation–model disagreement, and sign reversal.

Extraction fidelity. We annotated ten papers by hand from the same parsed text the extractor consumed, recording every study setting and own-result relationship and tagging each as a *main* finding (foregrounded in the abstract or conclusions, or a sign reversal, threshold, or stated condition) or a *supporting* one (robustness checks, significance statements, incidental magnitudes). Over the ten papers the pipeline recovers 111 of 178 main findings (0.624) and 122 of 265 total (0.460), at precision 0.984; per-paper values span 0.30 to 0.84 (Appendix B). Two findings matter more than the headline number. Recall is set by how finely the paper map cut the paper rather than by the claim stage: the correlation between claim scopes and main-claim recall is $r = 0.65$, and both single-scope papers sit at the bottom. And what is missed is not a random sample but the conditioning structure specifically — the lowest-recall paper loses its diurnal sign reversal and its $0.20 \text{ m}^3 \text{ m}^{-3}$ threshold, the deforestation paper the entire tropical-warming/boreal-cooling split, the Saharan paper the panel-efficiency case where the effect reverses. Each is a reversal in Appendix A. Staged extraction reliably captures headline effects and systematically drops the conditions that decide when they hold — at once the strongest argument for representing conditions explicitly and the clearest bound on how well any downstream retriever can do.

Table 1: Gold ScientificClaim recall on the eight in-corpus questions. Out-of-corpus transfer questions are excluded because a correct answer there is a conditioned hypothesis, not a retrieval hit.

Question	Probes	Gold	Hit	Recall
Q1 heterogeneous 14.4 km patches	single-paper mechanism	4	4	1.000
Q5 background wind 0 to 1 m s ⁻¹	single-variable sensitivity	5	5	1.000
Q7 deforestation cloud offset	multi-step mechanism chain	4	4	1.000
Q8 dry vs. wet soil rainfall	observation–model disagreement	3	3	1.000
Q4 patch-size threshold	threshold and scale dependence	3	1	0.333
Q9 afforestation sign	cross-paper sign reversal	6	2	0.333
Q2 Rondônia season and wind	seasonal conditioning	8	1	0.125
Q6 irrigation and extremes	cross-paper agreement	6	0	0.000
Mean				0.599

Query benchmark. Table 1 reports gold claim recall on the eight in-corpus questions. ClimateKG averages 0.599, with five questions fully recovered and two failures, neither a retrieval failure. On the Rondônia question every path comes from the correct paper, but the eight-path budget captures the seasonal branches and not the weakened-wind ones. On the irrigation question the gold claims rank 2nd, 5th, 6th and 7th of 200 candidates and are discarded before synthesis: the parser emitted two outcome variables as context facets, switching the gate on, and the gate dropped every claim whose applicability was *unknown* rather than mismatched — contradicting the pipeline’s own invariant that missing coverage is not mismatch.

Transfer behaviour. The out-of-corpus questions separate the systems more sharply than recall does. Asked about 10–15 km alternating irrigated and dry patches in semi-arid Rajasthan, GraphRAG blended US Midwest drought claims into its answer and advised that planners might “leverage” the dry-patch convection tendency — a placement recommendation for a region no source paper studied. Plain RAG returned generic irrigation caveats, never reaching the heterogeneity experiment that is the closest analogue. ClimateKG retrieved it, reported *spatial transfer* explicitly blocked, and surfaced the conflict between the user’s stated weak monsoon wind and the derived ERA5 JJA mean of 3.09 m s⁻¹.

4 Discussion and outlook

These answers constrain a scenario set rather than replace a model. The pipeline returns mechanism claims whose study conditions match the enriched query context, each carrying the patch scale, wind regime, season and soil-moisture state under which it was reported — exactly the parameters of a raster land-cover perturbation: which classes to convert, at what patch size, in what arrangement relative to prevailing wind. The output is a handful of candidate scenarios with provenance back to page-level text, ready to be harmonised into land-surface inputs [Hurt et al., 2020] and tested in a convection-permitting run [Skamarock et al., 2019].

The evidence is limited: ten questions, eighteen papers, one generation model, single runs. Gold claim sets were read off the extracted inventory, so the query benchmark measures whether a system finds the evidence the graph contains, not whether the graph contains the right evidence — the extraction audit addresses the latter, and the two must be read together. Setting counts are totals not matched pairs, RAGAS-style scoring is in progress, and the ground truth, though auditable against recorded evidence blocks, has not been independently re-annotated. Ten questions support no claim of general superiority; they support only that the three systems fail differently, and that one declines to transfer. Two fixes set the next step: finer scope segmentation in the paper map, the binding constraint on extraction recall, and a context gate treating unknown coverage as unknown rather than disqualifying — both serving the one property a literature-derived scenario needs, that the conditions under which an effect holds survive to the answer.

References

- Ramdane Alkama and Alessandro Cescatti. Biophysical climate impacts of recent changes in global forest cover. *Science*, 351(6273):600–604, 2016. doi: 10.1126/science.aac8083.
- Somnath Baidya Roy and Justin J. Traiteur. Impacts of wind farms on surface air temperatures. *Proceedings of the National Academy of Sciences*, 107(42):17899–17904, 2010. doi: 10.1073/pnas.1000493107.
- Nikolina Ban, Ccile Caillaud, Erika Coppola, Emanuela Pichelli, Stefan Sobolowski, et al. The first multi-model ensemble of regional climate simulations at kilometer-scale resolution, part i: evaluation of precipitation. *Climate Dynamics*, 57:275–302, 2021. doi: 10.1007/s00382-021-05708-w.
- Hylke E. Beck, Niklaus E. Zimmermann, Tim R. McVicar, Noemi Vergopolan, Alexis Berg, and Eric F. Wood. Present and future köppen-geiger climate classification maps at 1-km resolution. *Scientific Data*, 5, 2018. doi: 10.1038/sdata.2018.214.
- Qian Cao, Yaoyao Liu, Matei Georgescu, and Jianguo Wu. Impacts of landscape changes on local and regional climate: a systematic review. *Landscape Ecology*, 35(6):1269–1290, 2020. doi: 10.1007/s10980-020-01015-7.
- Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: a graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024.
- Partha Pratim Gogoi, V. Vinoj, D. Swain, G. Roberts, J. Dash, and S. Tripathy. Land use and land cover change effect on surface temperature over eastern india. *Scientific Reports*, 9, 2019. doi: 10.1038/s41598-019-45213-z.
- Hans Hersbach, Bill Bell, Paul Berrisford, Shoji Hirahara, et al. The era5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi: 10.1002/qj.3803.
- George C. Hurtt, Louise Chini, Ritvik Sahajpal, Steve Frolking, et al. Harmonization of global land use change and management for the period 850–2100 (luh2) for cmip6. *Geoscientific Model Development*, 13(11):5425–5464, 2020. doi: 10.5194/gmd-13-5425-2020.
- Jungmin M. Lee, Yunyan Zhang, and Stephen A. Klein. The effect of land surface heterogeneity and background wind on shallow cumulus clouds and the transition to deeper convection. *Journal of the Atmospheric Sciences*, 76(2):401–419, 2019. doi: 10.1175/JAS-D-18-0196.1.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- Yan Li, Eugenia Kalnay, Safa Motesharrei, Jorge Rivas, Fred Kucharski, Daniel Kirk-Davidoff, Eviatar Bach, and Ning Zeng. Climate model shows large-scale wind and solar farms in the sahara increase rain and vegetation. *Science*, 361(6406):1019–1022, 2018. doi: 10.1126/science.aar5629.
- Hao Luo, Johannes Quaas, and Yong Han. Decreased cloud cover partially offsets the cooling effects of surface albedo change due to deforestation. *Nature Communications*, 15, 2024. doi: 10.1038/s41467-024-51783-y.
- Nathaniel D. Mueller, Ethan E. Butler, Karen A. McKinnon, Andrew Rhines, Martin Tingley, N. Michele Holbrook, and Peter Huybers. Cooling of us midwest summer temperature extremes from cropland intensification. *Nature Climate Change*, 6:317–322, 2016. doi: 10.1038/nclimate2825.
- Roger A. Pielke Sr., Roni Avissar, Michael Raupach, A. Johannes Dolman, Xubin Zeng, and A. Scott Denning. An overview of regional land-use and land-cover impacts on rainfall. *Tellus B: Chemical and Physical Meteorology*, 59(3):587–601, 2007. doi: 10.1111/j.1600-0889.2007.00251.x.

208 Andreas F. Prein, Wolfgang Langhans, Giorgia Fosser, Andrew Ferrone, Nikolina Ban, Klaus Go-
209 ergen, Michael Keller, Merja Tölle, Oliver Gutjahr, Frauke Feser, Erwan Brisson, Stefan Kollet,
210 Juerg Schmidli, Nicole P. M. van Lipzig, and Ruby Leung. A review on regional convection-
211 permitting climate modeling: demonstrations, prospects, and challenges. *Reviews of Geophysics*,
212 53(2):323–361, 2015. doi: 10.1002/2014RG000475.

213 Jonas Schwaab, Edouard L. Davin, Peter Bebi, Anke Duguay-Tetzlaff, Lars T. Waser, Matthias
214 Haeni, and Ronny Meier. Increasing the broad-leaved tree fraction in european forests mitigates
215 hot temperature extremes. *Scientific Reports*, 10, 2020. doi: 10.1038/s41598-020-71055-1.

216 William C. Skamarock, Joseph B. Klemp, Jimy Dudhia, David O. Gill, Zhiquan Liu, Judith Berner,
217 Wei Wang, Jordan G. Powers, Michael G. Duda, Dale M. Barker, and Xiang-Yu Huang. A descrip-
218 tion of the advanced research wrf model version 4. *NCAR Technical Note NCAR/TN-556+STR*,
219 2019. doi: 10.5065/1dfh-6p97.

220 Abigail L. S. Swann, Inez Y. Fung, and John C. H. Chiang. Mid-latitude afforestation shifts general
221 circulation and tropical precipitation. *Proceedings of the National Academy of Sciences*, 109(3):
222 712–716, 2012. doi: 10.1073/pnas.1116706108.

223 Christopher M. Taylor, Richard A. M. de Jeu, Françoise Guichard, Phil P. Harris, and Wouter A.
224 Dorigo. Afternoon rain more likely over drier soils. *Nature*, 489(7416):423–426, 2012. doi:
225 10.1038/nature11377.

226 Adriaan J. Teuling, Christopher M. Taylor, Jan Fokke Meirink, Lieke A. Melsen, Diego G. Miralles,
227 Chiel C. van Heerwaarden, Robert Vautard, Annemiek I. Stegehuis, Gert-Jan Nabuurs, and Jordi
228 Vilà-Guerau de Arellano. Observational evidence for cloud cover enhancement over western
229 european forests. *Nature Communications*, 8, 2017. doi: 10.1038/ncomms14065.

230 Vincent A. Traag, Ludo Waltman, and Nees Jan van Eck. From louvain to leiden: guaranteeing
231 well-connected communities. *Scientific Reports*, 9, 2019. doi: 10.1038/s41598-019-41695-z.

232 Jingfeng Wang, Rafael L. Bras, and Elfatih A. B. Eltahir. The impact of observed deforestation on
233 the mesoscale distribution of rainfall and clouds in amazonia. *Journal of Applied Meteorology*,
234 39(7):1084–1096, 2000. doi: 10.1175/1520-0450(2000)039<1084:TIOODO>2.0.CO;2.

235 A Reported sign reversals in the benchmark corpus

236 Each row is a single intervention whose reported effect changes direction with a stated condition. An
237 answer that omits the condition column is wrong for one of the two branches. All rows are traceable
238 to extracted ScientificClaim identifiers in the released graph.

Intervention		Conditioning variable	Reported reversal
Soil anomaly	moisture	Time of day [Taylor et al., 2012]	Rain over drier soil 12:00–15:00; over wetter soil 21:00–03:00
Soil anomaly	moisture	Observation vs. model [Taylor et al., 2012]	Observations favour dry soils; parameterised models favour wet soils
Deforestation		Latitude [Luo et al., 2024]	Surface warming in the tropics; cooling in the boreal zone
Wind farm		Lapse rate [Baidya Roy and Traiteur, 2010]	Warming under stable air at night; cooling under unstable air by day
Solar farm		Panel efficiency [Li et al., 2018]	Rainfall increase at 15%; cooling and rainfall suppression at 45%
Deforestation (Rondônia)		Season and synoptic wind [Wang et al., 2000]	No effect in the rainy season; cloud only in the break period; localised rain in the dry season under weak wind
Afforestation		Soil moisture [Swann et al., 2012]	Warming where the water supply gap widens; ameliorated where moisture is abundant
Surface heterogeneity		Patch size [Lee et al., 2019]	Transition to deep convection above ~5 km; boundary-layer mixing below

B Full extraction audit

Ten papers annotated by hand from the exported SourceBlock text, using the counting rules of the extraction prompts so that reference and system objects are the same kind of thing. Study settings are one per complete combination of conditions the paper directly analyses. Claims are one per directed own-result relationship; only OWN_RESULT and AUTHORS_INTERPRETATION_OF_OWN_RESULT count, because the pipeline discards the other two evidence roles. *Main* findings are those foregrounded in the abstract or numbered conclusions, plus every sign reversal, threshold, and stated condition that changes when a main finding applies.

Paper	Ref. set.	Ext. set.	Main	Hit	Main rec.	All	All rec.
Rondônia deforestation [Wang et al., 2000]	16	20	32	27	0.844	50	0.660
European forest clouds [Teuling et al., 2017]	14	9	14	10	0.714	24	0.417
Saharan wind/solar farms [Li et al., 2018]	11	5	22	15	0.682	28	0.643
Odisha LULC warming [Gogoi et al., 2019]	16	9	17	11	0.647	27	0.444
Wind farm temperatures [Baidya Roy and Traiteur, 2010]	14	9	17	10	0.588	23	0.435
Deforestation cloud offset [Luo et al., 2024]	13	3	16	9	0.562	24	0.375
Mid-latitude afforestation [Swann et al., 2012]	13	3	18	10	0.556	28	0.357
Broad-leaved fraction [Schwaab et al., 2020]	14	10	15	8	0.533	23	0.391
US Midwest cooling [Mueller et al., 2016]	17	8	17	8	0.471	22	0.364
Afternoon rain, drier soils [Taylor et al., 2012]	10	3	10	3	0.300	16	0.188
Total	138	79	178	111	0.624	265	0.460

Precision over the ten papers is 0.984: of 127 extracted claims, 122 match a reference claim, two are cited background mis-assigned as own results, and three are duplicates of a single irrigation-cooling relationship.

C Benchmark corpus

The frozen 18-paper subset covers wind farms, deforestation and cloud cover, global and regional irrigation, mesoscale vegetation heterogeneity, soil-moisture–rainfall coupling, cropland intensification, large-scale renewable installations, forest cloud enhancement, mid-latitude afforestation, broad-leaved species composition, land-use change in South America, global greening, cover crops, observed Amazonian deforestation, LULC warming in eastern India, and land-surface heterogeneity with background wind. Full identifiers, titles, and per-paper object counts accompany the released graph.

D Reproduction

The shared corpus, the ClimateKG graph, the ten-question benchmark, the extraction ground truth, and the collation scripts are released with the code. The corpus is assembled by `scripts/build_three_system_corpus.py`, ClimateKG queries run through `scripts/run_three_system_queries.py`, the two baselines through `baselines/climatekg_comparison/run.py` with a `-think` flag that matches the ClimateKG synthesis budget, results are collated by `scripts/report_three_system_comparison.py`, and the extraction audit is recomputed by `scripts/audit_extraction_recall.py`. Every number in this paper is produced by one of those scripts from saved run artifacts.